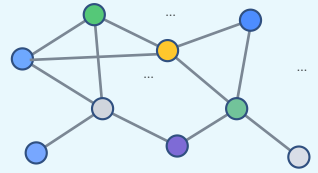
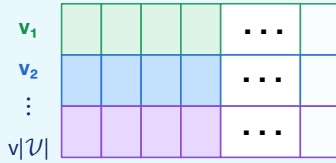


Input

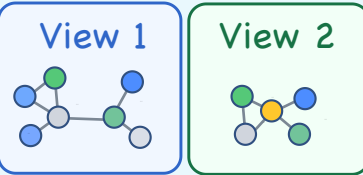
Graph Structure
 $\mathcal{G} = (\mathcal{V}, \mathcal{E})$



Node Features
 $\mathbf{X} \in \mathbb{R}^{|\mathcal{V}| \times F}$



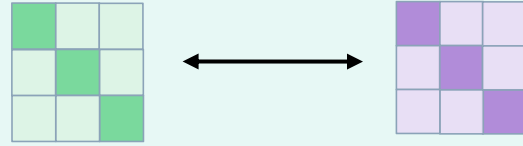
Data Augmentation



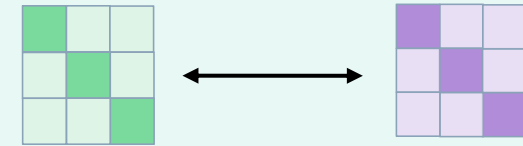
Generate two augmented graph views as the input

CDMD: Cross-Dimensional Mutual Distillation

Low-d_i Sim Matrix Max-d Sim Matrix



Low-d_j Sim Matrix Max-d Sim Matrix



: Similarity Matrix

$\longleftrightarrow$: Similarity/Alignment

Align similarity distributions with symmetric KL divergence

HPEM: Hard-Pair Evolutionary Mining

Small-d Prefix d_{i-1}
Representation Capacity

Capacity Used Limited

Node Difficulty (by confusion)



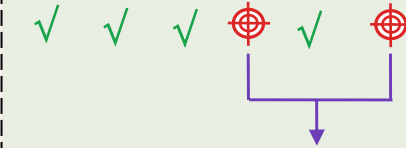
Hard Samples Discovered



Large-d Prefix d_i
Representation Capacity

Previous Capacity Extra Capacity

Focus on Hard Samples



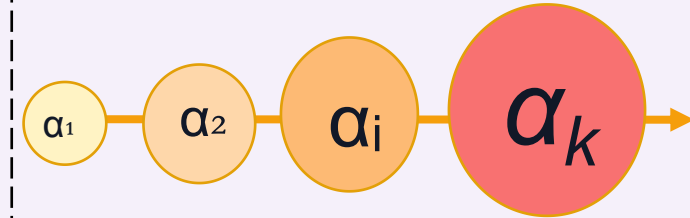
More optimization focus :
Higher Weight

Use small-d confusion to weight large-d InfoNCE loss

DALS: Dimension-Adaptive Loss Scheduling

$$\alpha_i = \exp \left(\lambda \frac{d_i}{d_K} \right)$$

Weights increase with dimension
: $\alpha_1 < \alpha_2 < \dots < \alpha_i < \dots < \alpha_k$



Weighted HPEM Loss Aggregation

$$\alpha_1 \cdot \mathcal{L}_{HPEM}^1 + \dots + \alpha_k \cdot \mathcal{L}_{HPEM}^k$$

Effect: Higher-dimensional prefixes receive larger weights

Joint Optimization:

$$\mathcal{L}_{MCNE} = \sum_{i=1}^K \alpha_i \mathcal{L}_{HPEM}^{(i)} + \mu \mathcal{L}_{CDMD}$$